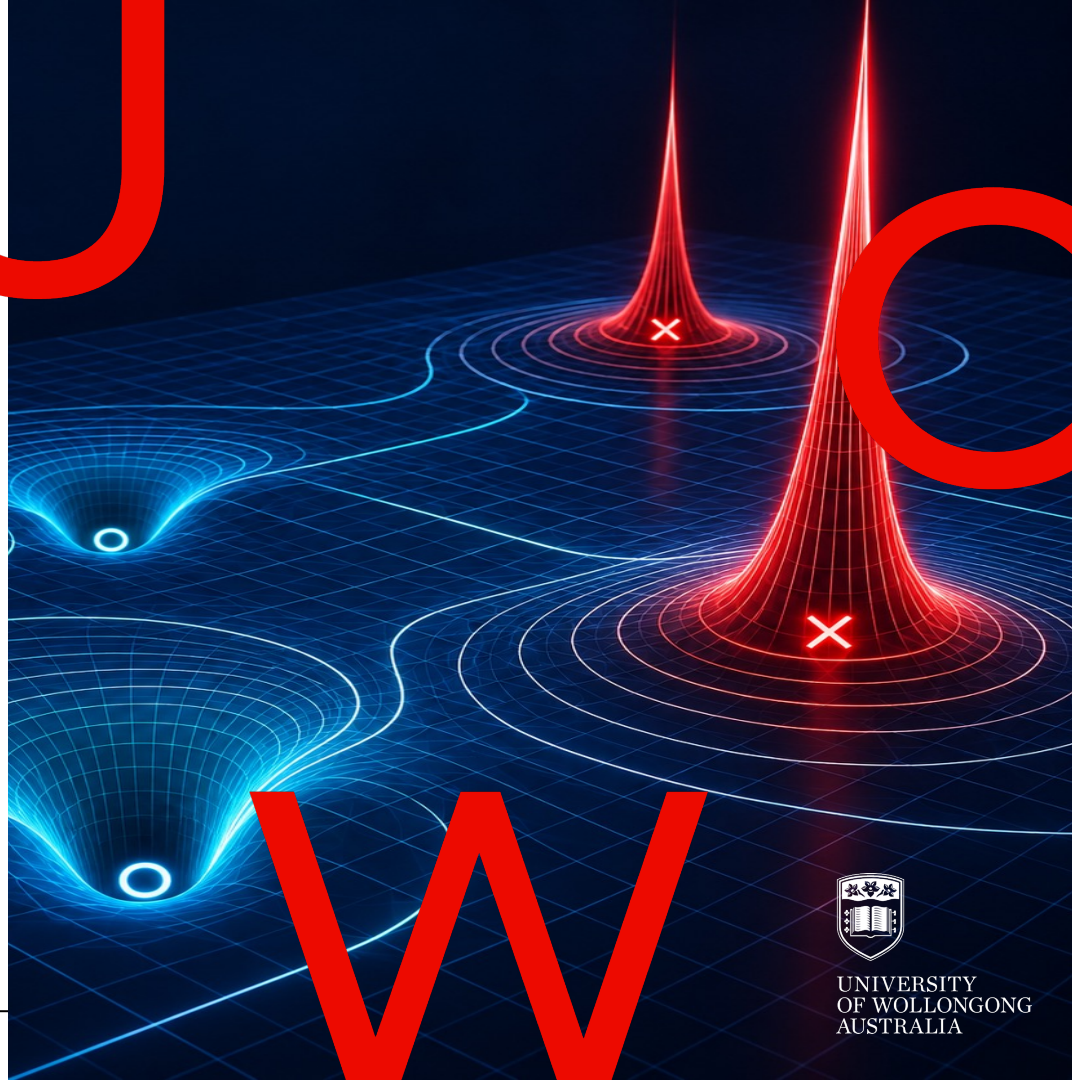
# MECH321 Dynamics of Engineering Systems

**Lecture 6**
**Physical Modelling and State Space**

Prof. Zengxi (Stephen) Pan

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PART 1

# Circuits and mechanical analogies

01 RC and RLC circuit models

02 Impedance and cascade loading

03 Mechanical–electrical analogy

# Modelling physical systems



| Mechanical | Electrical | Electromechanical | Fluid / thermal |
|---|---|---|---|
| Lectures 1–4 | $RC$, $RLC$ circuits | DC motor and robot joint | Tank, valve, sensor |

**Physical laws** → **ODE + order** → **Transfer function**

Newton, KVL, balances

Count independent stores

Zero-state input–output

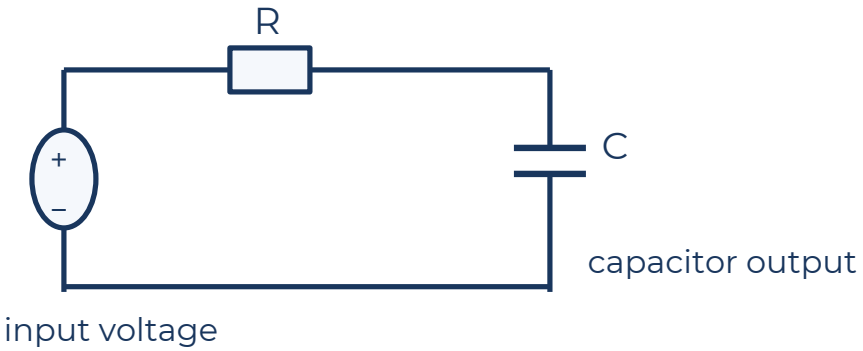
↘️ **State space**

Internal states + inputs

# RC circuit: current balance to transfer function

EXAMPLE

***Capacitor voltage is the single independent state***




**KVL → capacitor law → transfer function**

$$Ri + \frac{1}{C}\int i\,dt = e_i$$

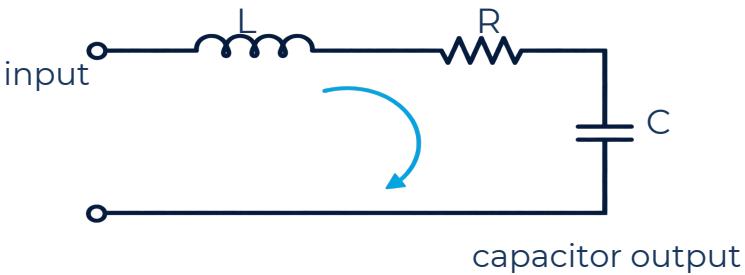
$$RI(s) + \frac{1}{Cs}I(s) = E_i(s)$$

$$\frac{E_o(s)}{E_i(s)} = \frac{(1/Cs)I(s)}{RI(s) + (1/Cs)I(s)} = \frac{1}{RCs + 1}$$

**If the capacitor initially holds charge, retain the initial-condition term; $G(s)$ alone does not describe that free response.**

# RLC Transfer Function

EXAMPLE




**MODEL → ODE → LAPLACE → OUTPUT**

$$L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt = e_i$$

$$LsI(s) + RI(s) + \frac{1}{Cs}I(s) = E_i(s)$$

$$\frac{E_o(s)}{E_i(s)} = \frac{(1/Cs)I(s)}{LsI(s) + RI(s) + (1/Cs)I(s)} = \frac{1}{LCs^2 + RCs + 1}$$

1 Series current through $R$, $L$ and C. KVL; capacitor-voltage output.
3 Zero-state transform: integral $1/s$; derivative $s$.
4 Multiply by $Cs$. Independent $L$ and $C$ stores: $s^2$ denominator.

**Denominator coefficients: LC, $RC$, 1.**

# Worked circuit: series RLC

EXAMPLE

$R = 4\ \Omega, L = 1\ \text{H}, C = 0.25\ \text{F}$

Input $v(t)$, output capacitor voltage $v_C(t)$.

**KVL around the loop:**

$$\frac{Ldi}{dt} + Ri + v_C = v$$

The capacitor links current to the output:

$$i = \frac{Cdv_C}{dt}$$

**Eliminate $i$ and collect**

$$LC\ddot{v}_C + RC\dot{v}_C + v_C = v$$

$$0.25\ddot{v}_C + \dot{v}_C + v_C = v$$

$$\frac{V_C(s)}{V(s)} = \frac{1}{0.25s^2 + s + 1}$$

$\omega_n = 2$ rad/s, $\zeta = 1 \rightarrow$ **critically damped**

Read $\zeta$ and $\omega_n$ using the standard form.

# Complex Impedance

CORE

***Use element impedances for transfer functions and sinusoidal gain and phase***

**Zero initial conditions: each element obeys $V(s) = Z(s)I(s)$.**

| Element | Time-domain law | Impedance $Z(s)$ |
|---|---|---|
| **Resistor** | $v = Ri$ | $Z_R = R$ |
| **Inductor** | $v = L\frac{di}{dt}$ | $Z_L = sL$ |
| **Capacitor** | $i = C\frac{dv}{dt}$ | $Z_C = \frac{1}{sC}$ |

**For a series loop, add impedances. Select the output using a voltage divider.**
Nonzero initial energy requires additional source terms.

# Worked impedance solution: series RLC

EXAMPLE

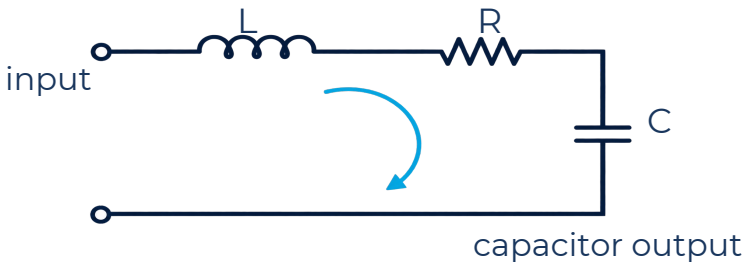
**Question Series $R$–$L$–$C$; source $u(t)$, output capacitor voltage $v_C(t)$.**

The same circuit and values as the preceding KVL example: $R = 4$ Ω, $L = 1$ H, $C = 0.25$ F.

$$Z_R = R, \qquad Z_L = sL, \qquad Z_C = \frac{1}{sC}$$

$$G(s) = \frac{Z_C}{Z_L + Z_R + Z_C} = \frac{1}{LCs^2 + RCs + 1}$$

$$G(s) = \frac{1}{0.25s^2 + s + 1} = \frac{4}{(s+2)^2}$$




**Check: two stores give two poles. The repeated pole at −2 matches the KVL derivation.**

# Mechanical–electrical analogy



| Role | Mechanical | Electrical | Physical meaning |
|---|---|---|---|
| **Effort / flow** | force / velocity | voltage / current | power-conjugate pair |
| **Inertia** | mass | inductor | energy storage |
| **Dissipation** | viscous damper | resistor | energy loss |
| **Elasticity** | spring | capacitor | stiffness maps to reciprocal capacitance |

**Force–voltage analogy**

Matching balance equations; different physical units. Use zero initial displacement and charge.

PART 2

# DC motor and first-order models

01 DC motor coupling

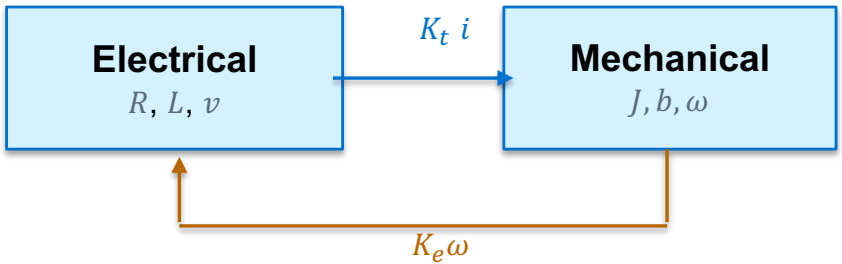
02 Motor reduction

03 Thermal extension

# The DC motor

CORE

***Two subsystems, coupled in both directions***




**The three equations**

$$\frac{Ldi}{dt} + Ri + K_e\omega = v$$

$$T = K_t i$$

$$\frac{Jd\omega}{dt} + b\omega = T$$

**Eliminate $i$ and $T$:**

$$\frac{\Omega(s)}{V(s)} = \frac{K_t}{(Ls+R)(Js+b) + K_tK_e}$$

Back EMF $K_e\omega$ couples speed to the electrical balance; $K_tK_e$ appears in the denominator.

# DC motor: voltage and opposing load

CORE

***Eliminate current once, then separate the two speed-response paths***

1 Positive load opposes positive rotation (slide 12).

$$J\frac{d\omega}{dt} + b\omega = K_t i - \tau_l(t)$$

2 Zero initial current and speed: transform both balances.

$$(Ls + R)I(s) + K_e\Omega(s) = V(s)$$

$$(Js + b)\Omega(s) = K_t I(s) - \mathcal{T}_l(s)$$

3 Solve KVL for current. Substitute into torque balance, then multiply by the electrical factor.

$$I(s) = \frac{V(s) - K_e\Omega(s)}{Ls + R}$$

$$D(s) = (Ls + R)(Js + b) + K_t K_e$$

4 Zero the other input. Same poles; different numerators and signs.

$$D(s)\Omega(s) = K_t V(s) - (Ls + R)\mathcal{T}_l(s)$$

$$G_v(s) = \frac{K_t}{D(s)} \qquad G_\tau(s) = -\frac{Ls + R}{D(s)}$$

# Reduced DC motor model

CORE

***Notation: K=Kt, Kb=Ke, La=L, Ra=R and Ea=V from the preceding motor model.***

1 Current and speed states. Electrical scale $\tau_e = L_a/R_a$; loaded mechanical scale $\frac{J}{b+KK_b/R_a}$.
2 Quasi-steady current requires $\tau_e$ much shorter than motion and input time scales.
3 Set $L_a \approx 0$; substitute algebraic current into torque balance: first-order speed.
Position integrates speed: one additional pole at zero.

**FULL MODEL**

$$\frac{\Theta(s)}{E_a(s)} = \frac{K}{s[L_aJs^2 + (L_ab + R_aJ)s + R_ab + KK_b]}$$

**REDUCED POSITION TRANSFER FUNCTION**

$$\frac{\Theta(s)}{E_a(s)} = \frac{K}{s(R_aJs + R_ab + KK_b)}$$

**STANDARD REDUCED POSITION FORM**

$$\frac{\Theta(s)}{E_a(s)} = \frac{K_m}{s(T_ms + 1)}$$

**Boundary: use the full model for rapid voltage changes, current-control transients or startup.**

# The universal first-order model



***In the tank row, C is cross-sectional area in square metres***

$$\tau\dot{y} + y = Ku \quad G(s) = \frac{K}{\tau s + 1}$$

$\tau$ — time constant, seconds $K$ — DC gain, output units per input unit

| System | Equation | Time constant |
|---|---|---|
| **RC circuit** | $RC\dot{v}_C + v_C = v_{\text{in}}$ | $\tau = RC$ |
| **Tank level** | $RC\dot{h} + h = Rq_{\text{in}}$ | $\tau = RC$ |
| **Thermometer** | $RC\dot{\theta} + \theta = \theta_a$ | $\tau = RC$ |
| **Motor, $L$ neglected** | $\frac{J}{b_{\text{eff}}}\dot{\omega} + \omega = Kv$ | $\tau = \frac{J}{b_{\text{eff}}}$ |

Motor row (armature inductance neglected):

$$b_{\text{eff}} = b + \frac{K_t K_e}{R}, \quad K = \frac{K_t}{Rb_{\text{eff}}}$$

# Thermometer Model

EXAMPLE

Heat balance: storage rate = incoming heat.

2 Use $q = \frac{\theta_b - \theta}{R}$ and $C\frac{d\theta}{dt} = q$.

3 Heat balance: $RC\frac{d\theta}{dt} + \theta = \theta_b$.

4 Under zero initial deviation, transform and take output/input: $\tau = RC$, gain=1.

$$C\frac{d\theta}{dt} = q = \frac{\theta_b - \theta}{R} \quad \text{or} \quad RC\frac{d\theta}{dt} + \theta = \theta_b$$

$$RCs\Theta(s) + \Theta(s) = \Theta_b(s)$$

$$\frac{\Theta(s)}{\Theta_b(s)} = \frac{1}{RCs + 1}$$

**For an initially hot thermometer, include $\theta(0)$ in the Laplace derivative. The transfer function is the zero-state input-output relation.**

# Worked: the thermal sensor

EXAMPLE

$12\dot{\theta} + \theta = \theta_a$

$\tau = 12$ s. Ambient steps from 20 °C to 80 °C; the sensor reads 20 °C at $t = 0$.

**Find the temperature response, the reading at 12 s and the approximate 2% settling time.**

$\Delta\theta = 60°$ C

**Answer, then check**

$\theta(t) = 80 - 60e^{-\frac{t}{12}}°$ C

At $t = \tau = 12$ s: 63.2 % of 60 °C is 37.9 °C

$\theta(12) = 20 + 37.9 = 57.9°$ C✓

**2 % settling** $\approx 4\tau = 48$ s

PART 3

# State space and system modes

01 State equation and solution

02 Eigenvalues and modal directions

03 Transfer-function relationship

# State model and transfer-function view

CORE

## GIVEN STATE MODEL

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \quad 0]x, \quad x(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

State 1 is the output. State 2 is its rate.
The input is a unit step and the initial state is zero.

## TRANSFER-FUNCTION VIEW

$$G(s) = \frac{Y(s)}{U(s)}$$

One selected input–output path
Poles and zeros
Zero initial conditions

## STATE-SPACE VIEW

Internal state trajectory
Initial conditions retained
Several inputs and outputs

# Eigenvalues, eigenvectors and natural modes

CORE

***The poles give decay rates; eigenvectors show how the states move together***

$$\det(\lambda I - A) = \lambda^2 + 3\lambda + 2$$

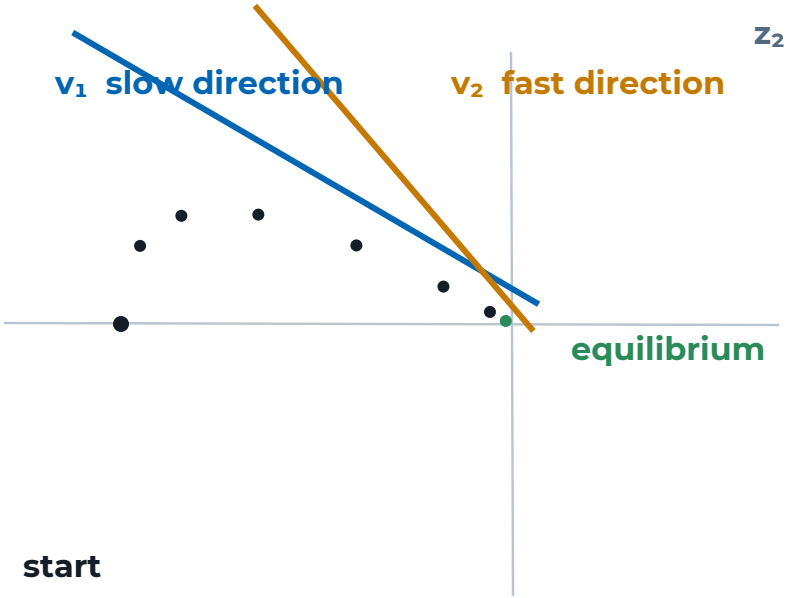
$$(\lambda + 1)(\lambda + 2) = 0$$

$$\lambda_1 = -1, \quad \lambda_2 = -2$$

$$v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \qquad v_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$x_h(t) = c_1 v_1 e^{-t} + c_2 v_2 e^{-2t}$$

Each eigenvalue sets a decay rate.
Each eigenvector sets a fixed state direction.
The slower mode controls the late response.




First-state deviation from its steady value

The state points approach equilibrium as the fast mode fades and the slow mode remains.

# State response and its two causes

CORE

***Initial energy and external input appear as separate terms***

## GENERAL SOLUTION

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau$$

## INITIAL-STATE RESPONSE

$$e^{At}x(0)$$

Present with zero input.
Carries stored-state information.

## INPUT-DRIVEN RESPONSE

Present from the zero initial state.
This is the transfer-function path.

$$u = \text{constant:} \quad x(t) = x_{ss} + e^{At}[x(0) - x_{ss}]$$

# Unit-step response

EXAMPLE

***Shift to the equilibrium, determine the modal coefficients and recover both states***

## SETUP AND MODAL COEFFICIENTS

$$x_{ss} = -A^{-1}B = \begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix}$$

$$z(t) = x(t) - x_{ss}, \quad z(0) = \begin{bmatrix} -\frac{1}{2} \\ 0 \end{bmatrix}$$

$$z(t) = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{-2t}$$

$$c_1 = -1, \quad c_2 = \frac{1}{2}$$

## STATE TRAJECTORY

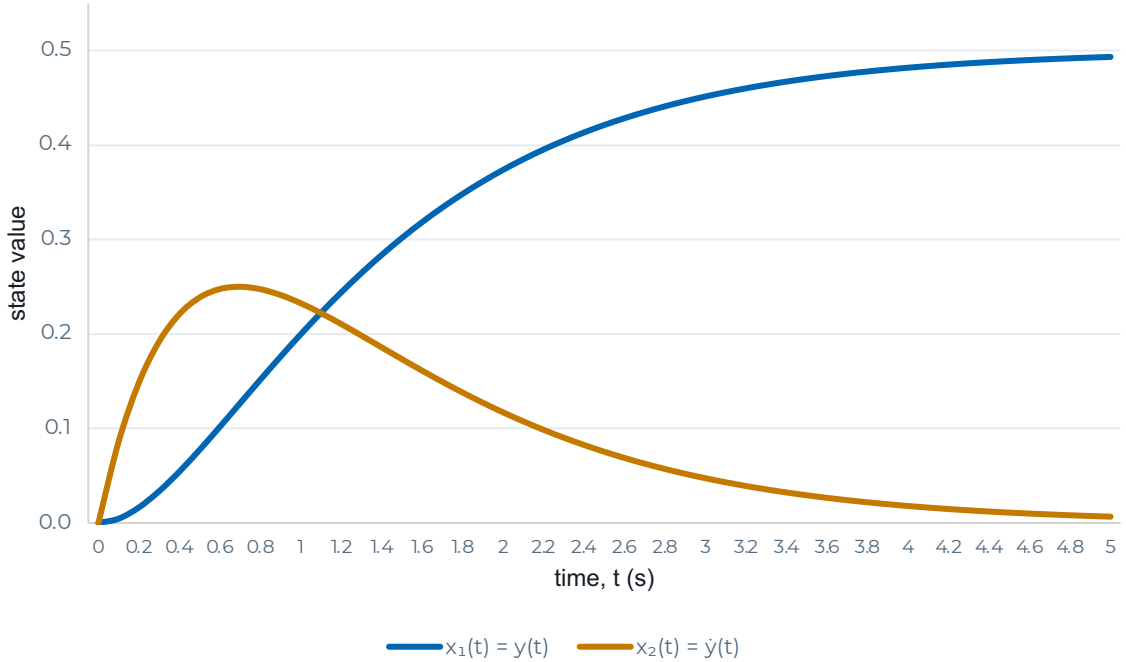
$$x_1(t) = \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}$$

$$x_2(t) = e^{-t} - e^{-2t}$$

**Checks: the response starts from the zero initial state; the final state is the equilibrium calculated above.**

# State trajectories






## READ THE CURVES

The first state rises monotonically to 0.5.

The second state peaks at 0.693 seconds, then returns to zero.

# Transfer function from the matrices

CORE

***The characteristic polynomial appears again in the denominator***

## GENERAL RELATION

$$X(s) = (sI - A)^{-1}BU(s)$$

$$G(s) = C(sI - A)^{-1}B + D$$

Set the initial state to zero.
The matrices then describe the chosen input–output channel.

## TWO-STATE EXAMPLE

$$sI - A = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

$$\det(sI - A) = s^2 + 3s + 2$$

$$G(s) = \frac{1}{s^2 + 3s + 2}$$

**For a minimal realization, transfer-function poles equal the state-matrix eigenvalues, and dimension equals order. This example has no cancellation.**

# State coordinates and multiple signals



## STATE COORDINATES

$$x = \begin{bmatrix} y \\ \dot{y} \end{bmatrix}$$

**Canonical states**

$$x_p = \begin{bmatrix} i \\ \omega \end{bmatrix}$$

**Physical motor states**

$$x_p = Tx \quad \Longrightarrow \quad A_p = TAT^{-1}$$

A nonsingular coordinate change alters the matrices but not the external transfer function.

For a minimal model, state dimension equals transfer-function order.

## MULTIPLE INPUTS AND OUTPUTS

$$u = \begin{bmatrix} v \\ \tau_L \end{bmatrix} \qquad y = \begin{bmatrix} i \\ \omega \end{bmatrix}$$

$$B \in \mathbb{R}^{n \times m}, \quad C \in \mathbb{R}^{p \times n}$$

B connects inputs to state derivatives; C selects measured outputs.

$$G(s) = \left[ G_{ij}(s) \right]$$

**Multiple channels form a transfer-function matrix.**

# Transfer function and state space

CORE

## TRANSFER FUNCTION

Selected input–output path, with zero initial state

Poles, zeros and frequency response for loop shaping

## STATE SPACE

Internal trajectories, with initial conditions explicit

Several inputs and outputs in one model

Supports state estimation and feedback

## OPTIONAL EXTENSION: CONTROLLABILITY AND OBSERVABILITY

**Can the input move every state direction?**

$$\mathbf{C} = [B \;\; AB \;\; \cdots \;\; A^{n-1}B]$$

$$\operatorname{rank}(\mathbf{C}) = n$$

**Can the outputs reveal every state direction?**

$$\mathbf{O} = [C^T \quad (CA)^T \quad \cdots \quad (CA^{n-1})^T]^T$$

$$\operatorname{rank}(\mathbf{O}) = n$$


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